Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Convex Perturbation Paper - My Response to Your Latest Comments

Benjamin D. Summers <bsummers@adagiollc.com> Sun, Sep 27, 2026 at 10:18 PM
To: Ezequiel Boero <ezequiel.boero@adagioinstitute.org>
Cc: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Hi Ezequiel,

Thank you. Both points are in; revision 4 is on SSRN under the same DOI (https://dx.doi.org/10.2139/ssrn.6812461).

1. Scale-level witness. Proposition 4.10 now has two parts. Part (i) is the previous statement. Part (ii) is your construction: for divisible Γ_a and h > 0, Γ_0 = (Γ_a + hΓ_ref)/(1 + h), which is (14) at g* = −h, is a two-time family with X = −hK at every triple, D_CK(δt;Γ_0) = hM_*(δt), and D_CK(δt;Γ_g) = (1 − g)(h + g)M_*(δt) = E(g) identically, so g_first = g_term = g_1 at the scale level. The proof uses that the identity (6) is an algebraic expansion valid for every real g, and evaluates it at g = −h, where Γ_g = Γ_a. A paragraph after the proof reads (i) and (ii) as one formula with Γ_a on either side of Γ_0 along the affine line through Γ_ref, and notes that the entrywise condition is automatic for g* < 0. Remark 4.5 now cites 4.10(ii) for the attainment at the scale level and labels the λ = 0.35 configuration pointwise; the introduction, the outline, and the lead of Section 4.3 say the same. The abstract is unchanged; its claim now has the witness it needed.
2. Instance. With the same Glauber pair and h = 0.6: a(τ) = (e^{−2τ} − e^{−4τ})/(1 + h), M_*(δt) = 1/[16(1 + h)²] = 0.024414 for δt ≥ ln 2, D_CK(δt;Γ_0) = 0.014648, and D_CK(δt;Γ_g) = 0.024414(1 − g)(0.6 + g), which rises on [0, 0.2] before it falls (a scale-level instance of the initial rise in Remark 3.3). At ε = 0.01 the numbers come out exact: g_first = g_term = g_1 = 0.68 and g_0 = 0.744. My checks on 3,000 random triples give max ‖X + hK‖ = $3.0\cdot10^{-16}$ and max |D − E| = $6.9\cdot10^{-18}$.
3. Remark 6.2 now reads "for 0 < g < 1 it reduces to the fixed weight g exactly when π^Y_s = π^Z_s for all s (at g = 0 and g = 1 the weight is constant whatever the laws)."
4. The 1/64. You are right that it is not accidental; the manuscript now says the value is the same for every g* ∈ (0, ½], the range in which Γ_0 is stochastic.

Numbering: no theorem, proposition, remark, or section number moved. The new display (16) shifts the equation numbers of Sections 5 and 6 by one, so the state-dependent class is now (19) and its bound (21); the cross-references in the manuscript are automatic.

Thank you for the reviews.

Ben

Benjamin D. Summers
Managing Director
ADAGIO GROUP
MOBILE +1.713.775.1011 | **OFFICE** +1.305.433.1823 | **FAX** +1 239.932.5558
bsummers@adagiollc.com | adagio.capital

On Sun, Sep 27, 2026 at 6:15 PM Ezequiel Boero <ezequiel.boero@adagioinstitute.org> wrote:

> Hi Ben,
>
> All four wording points and both results are in.
> One amendment remains but overall I would consider the review of this paper closed with essentially nothing outstanding.
>
> **What checks out.** Proposition 4.10 I re-derived in both directions; it is right. Remark 4.5's attainment condition is correct as a conditional statement.
> The four wording points are in as described.
>
> The re-entry point at ε = 0.01 is 0.4882, not the 0.4885 I reported. I had computed it with an M* taken from an unresolved grid of triples.
> The manuscript prints the right figure.